

Questions for Elementary School Solar Group

Topic: Elementary School Solar Meeting

Time: Jan 31, 2025 09:30 AM Eastern Time (US and Canada)

<https://us02web.zoom.us/j/81036738035?pwd=C684PKosgqunSW4v52ntpS3zu0kHEo.1>

Meeting ID: 810 3673 8035

Passcode: 048654

1. Confirm age, size, condition, status, and remaining life of existing generators serving the property (building and well). Does the generator provide electric backup to the entire school when power is out or just critical circuits at the school? If only critical circuits, which ones?
2. Confirm current status as an emergency shelter and town's desire/plan for future use as a shelter.
 - a. According to Todd Alexander, Emergency Management Director, the Elementary school currently would be categorized as a personal care site which only provides limited services such as warming and cooling assistance, limited food and water, restrooms, and charging stations during business hours, no overnight. For overnight shelters residents would need to go to the regional shelter facility at Smith Vocational. In what situations would we use it as such? Which of these services would we provide? Who would staff it?
 - b. Is this sufficient to meet town needs? Is there a desire to designate this site as an overnight shelter or provide more services as a personal care site?
 - c. Is the current equipment satisfactory to provide the desired services in a or b? If not, what improvements would be necessary?
 - d. Is there currently air conditioning in the gym/cafeteria? If not, how would the school be used as a cooling center?
 - e. Who makes decisions regarding maintenance or repair of current equipment to ensure emergency readiness?
3. To price out a battery with microgrid SolectEnergy needs the model and serial number of the Automatic Transfer Switch (ATS) associated with the generator. Can someone provide that information?
 - a. Is there one place where all system information is kept? Unit type, installation information, maintenance logs, etc.
4. In order to appropriately size the solar array, we need to better understand current and future energy needs.
 - a. Who has confirmed the heat pumps cannot be used in winter? Have we gotten detailed information on why? With additional air sealing could they be used?
 - b. No immediate plans to make energy upgrades to the gymnasium, but is this being considered? Does the school committee consider these types of upgrades or another entity?
 - c. In order to confirm the PV system size, Solect needs to know if the school will or will not be adding mini splits or some other kind of heat pump. The current system design covers around 109% of the school's current consumption at its high points over the past 2 years. Solect can increase the system size if it's warranted.

5. Confirm final location and type of solar panels. The School Committee supported ground mounted in the location as proposed initially by SolectEnergy (in the back beyond the play area).
 - a. They also asked for Solect to evaluate the feasibility of an alternate site along the northwest side of the building near the front parking lot. During the site visit on 1/15/25, Solect did view this location and recommends keeping the original site because of its proximity and ease of access in connecting the solar panels to the electric box entering the building on the southeast side of the building. It is also a larger space and can accommodate expansion towards Hathaway Road if necessary.
 - b. Confirm a ground mounted solar panel system in the original location so that Solect can begin their system design.
6. Confirm which battery backup system is preferred and should be designed for – either a standard solar powered battery backup system or a microgrid battery backup system.
 - a. MVP grant program staff have confirmed that MVP would consider an application for a solar system that includes a standard solar powered battery backup system that is not a microgrid even though it will have limited hours of battery charge and may need to use a fossil fuel powered generator for longer power outages. The goal is to incrementally move towards a net 0 building.
 - i. MVP would not pay for any costs related to a generator and those costs would also not be allowable match, but this year MVP is likely eliminating the match requirement for small towns. (great news)
 - ii. MVP may pay for heat pumps or geothermal systems but there is no precedence. It complicates the application, but if MVP decided not to fund this portion of the project, they would remove it from the scope of work rather than deny the project (if it is a competitive project otherwise). It also could be a separate project for the future.
7. MVP would be seeking partnerships for all applications and suggests the school as a partner in student engagement. Options could include:
 - a. Student curriculum on solar energy and tracking the energy generation of the system on site.
 - b. Partner with the Grinspoon Foundation Cooler Communities Program again to assist with curriculum for this age group of students (we did this last time at the high school). The grant would pay for this assistance but teachers and curriculum development staff would need to be involved.
 - c. Partner with the COA to host one or more joint senior/student presentations on the solar system and the benefits of solar energy.
 - d. Pollinator friendly plantings in and around the solar panels – perhaps the students can be involved in planning and maintaining plantings – would need to get discuss feasibility of this with Solect from a safety and functional perspective.